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July 20, 2026

Editors-in-Chief
Journal of Vibration and Control
SAGE Publications

Dear Editors,

Please consider the enclosed manuscript, *"Solution-Operator Semi-Discretization: General-Order Multiplication-Free Stability Analysis of Time-Periodic Delayed Systems and the Accuracy–Sparsity Trade-Off"*, for publication in the *Journal of Vibration and Control* as an original research article.

The manuscript is the direct continuation of my recent JVC article on the multiplication-free semi-discretization method (MFSD), which established the linear-time-complexity representation of the one-period solution operator of time-periodic delay differential equations. The present work answers the natural follow-up question: once the analysis is written in that multiplication-free form, *which integrator should be placed inside each step?* I show that the per-step integrator is a free design choice — any explicit Runge–Kutta or implicit collocation-type scheme embeds through its Butcher tableau — and that the resulting accuracy–sparsity continuum possesses a quantifiable, problem-dependent CPU-time optimum (a "sweet spot") at moderate order and many steps. A Shannon-type sampling bound turns this observation into an a priori recipe for selecting the stage number and step count, validated on four benchmark systems including machining applications, and demonstrated end-to-end by high-resolution stability charts computed in seconds. All methods and every figure are reproducible through the open-source Julia package SOSD.jl.

The paper is deliberately explicit about its relation to prior art: neither the multiplication-free assembly (my previous JVC article) nor collocation itself is claimed as new; the contributions are the general-order embedding, the fair order-versus-step benchmarking methodology, the measured accuracy–sparsity trade-off, and the resulting practical guidance. I believe this continuity makes JVC the natural venue, and the same readership that received the predecessor article will find the sequel directly useful.

The manuscript is not under consideration elsewhere, and I have no conflicts of interest to declare. The research was supported by the Hungarian Scientific Research Fund (grant number NKFIH-AG-152125).

Thank you for your consideration.

Sincerely,

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